SOFTWARE ENGINEERING LAB

PROJECT REPORT

BCSE301P

L15 + L16

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23BCE5138

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1. Problem Statement

Citizens currently lack a unified, intelligent, and real-time platform to report public safety incidents such as theft, harassment, fire outbreaks, road hazards, and medical emergencies. Existing systems are fragmented, slow, and do not effectively capture geo-tagged evidence or prioritize incidents based on severity.

Emergency response mechanisms often suffer from delayed information flow and absence of automated risk assessment.

The objective of **S.H.I.E.L.D** is to develop a structured, AI-assisted public safety reporting system that enables:

- Real-time incident reporting
- Geo-location tagging
- Automated severity computation
- Visual mapping of incidents
- Conditional alert generation

The system aims to provide a scalable and modular foundation that can later integrate AI-driven intelligence components.

2. Objectives

The primary objectives of the project are:

1. To design and implement a secure incident reporting platform.
2. To capture and store geo-tagged incident data.
3. To compute severity scores for incidents based on contextual logic.
4. To visualize incidents on a real-time map interface.
5. To generate alerts for high-severity incidents.
6. To architect the system in a modular way that supports AI integration (NLP, image analysis, clustering).
7. To follow a structured Agile Scrum development process.

3. Methodology/Approach

The project follows the **Agile Scrum model** with time-boxed sprints.

Development Strategy

- Sprint-based incremental development
- Early delivery of functional modules
- Core system stabilization before AI integration
- Modular backend services for scalability

Technical Architecture

The system follows a service-oriented architecture consisting of:

- Frontend Interface (User Interaction)
- Backend API Layer (Business Logic)
- Database Layer (Persistent Storage)
- AI Service Layer

Technologies Used (Current Phase)

- Frontend
- Backend
- Database: MongoDB
- Map Integration: Leaflet

4. Work Completed So Far (50% Implementation)

At the 50% milestone, the core functional foundation of the system has been successfully implemented.

4.1 Project Setup & Architecture

- Backend project structure established
- API routing configured
- Database connection configured
- Environment configuration completed
- Frontend-backend communication verified

The system runs successfully in a local development environment.

4.2 User Authentication Module

- User registration implemented
- User login implemented
- Session/JWT authentication mechanism
- Basic trustScore field integrated in user model

Users must be authenticated before submitting incidents.

4.3 Incident Reporting Module

Users can:

- Select incident type
- Enter description
- Auto-capture GPS location
- Submit report

All incident data is validated and stored in the database.

Incident entity includes:

- Incident ID
- Type
- Description
- Timestamp
- Location coordinates

- Severity score
- Status

4.5 Database Schema Implementation

The following core tables are implemented:

- Users
- Incidents
- Location (embedded or relational)
- Alerts (basic structure)

Data persistence and retrieval have been verified.

4.6 Live Map Visualization

- All stored incidents are displayed on a map.
- Markers are color-coded based on severity.
- Incident details can be viewed interactively.

This confirms successful integration of:

- Backend APIs
- Database
- Frontend rendering

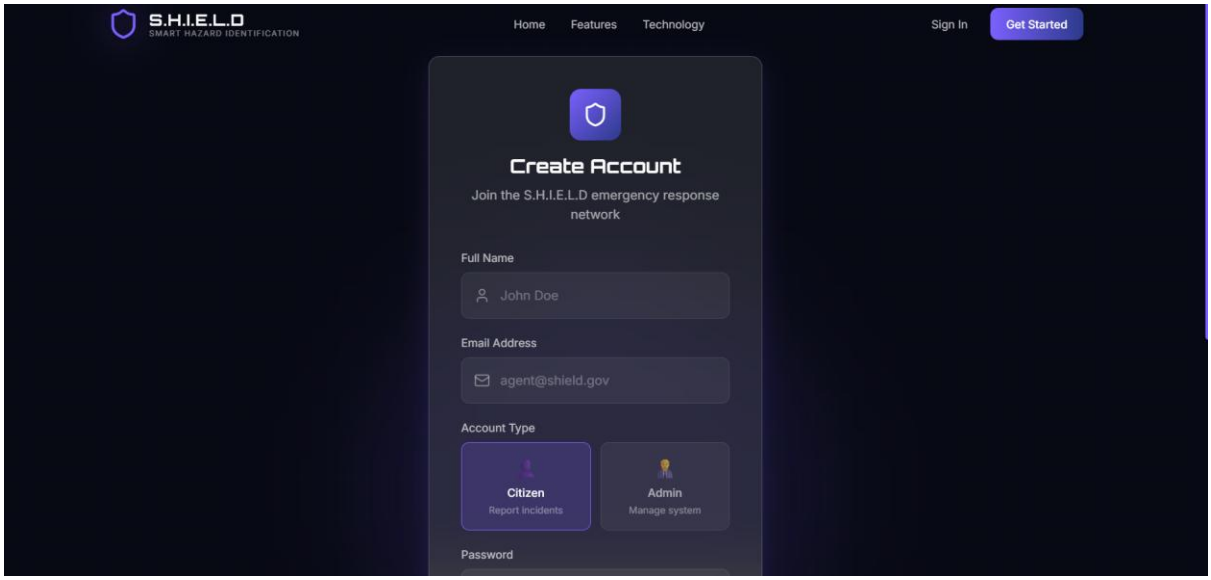
4.7 Basic Alert Trigger

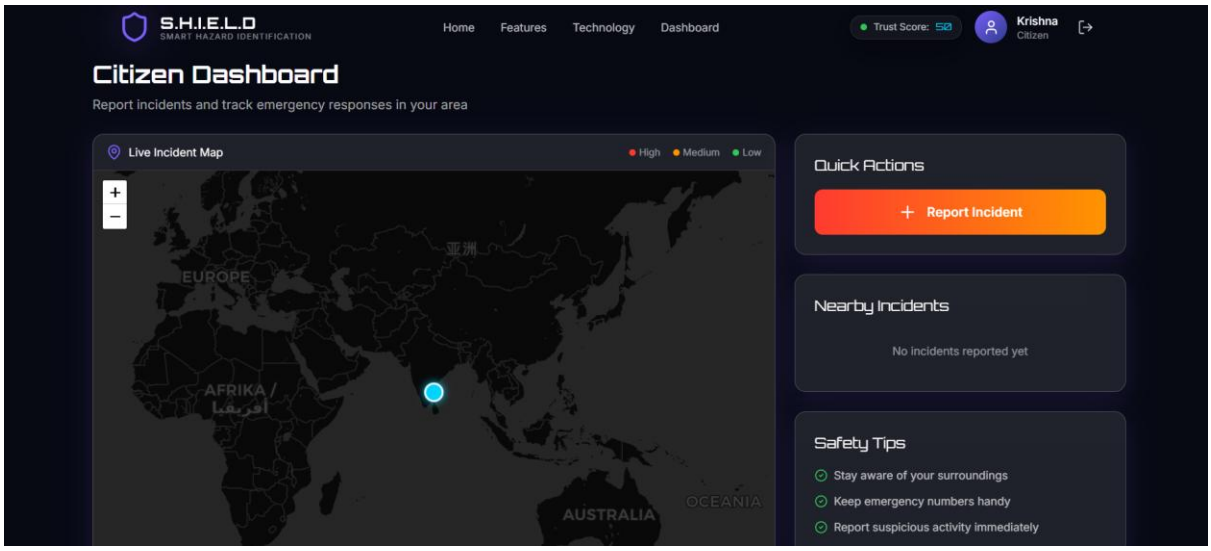
If severity exceeds a defined threshold:

- The system flags the incident.
- The admin dashboard displays high-severity incidents.

External notifications are planned for Phase 2.

5. Screenshots / Proof of Implementation



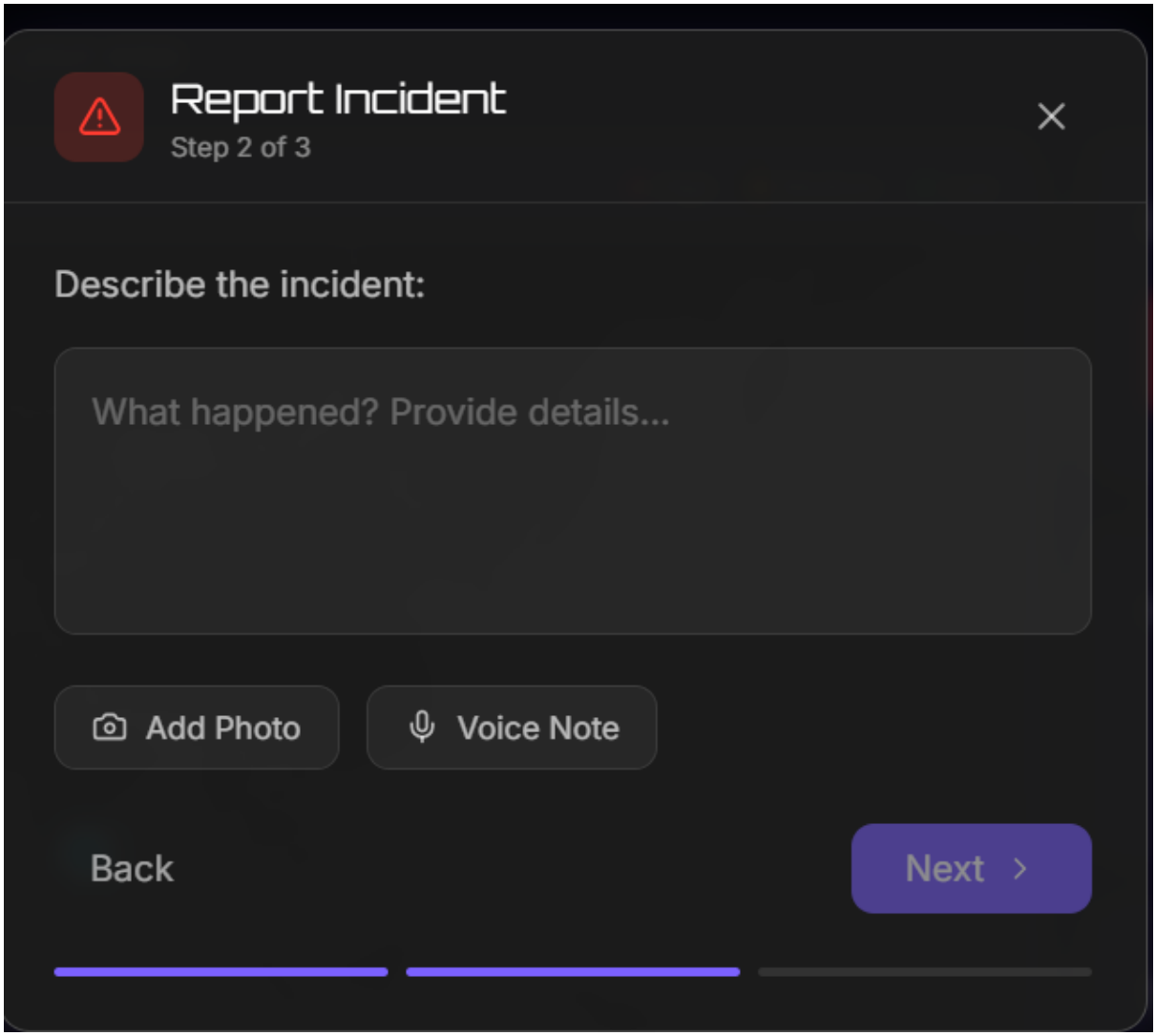


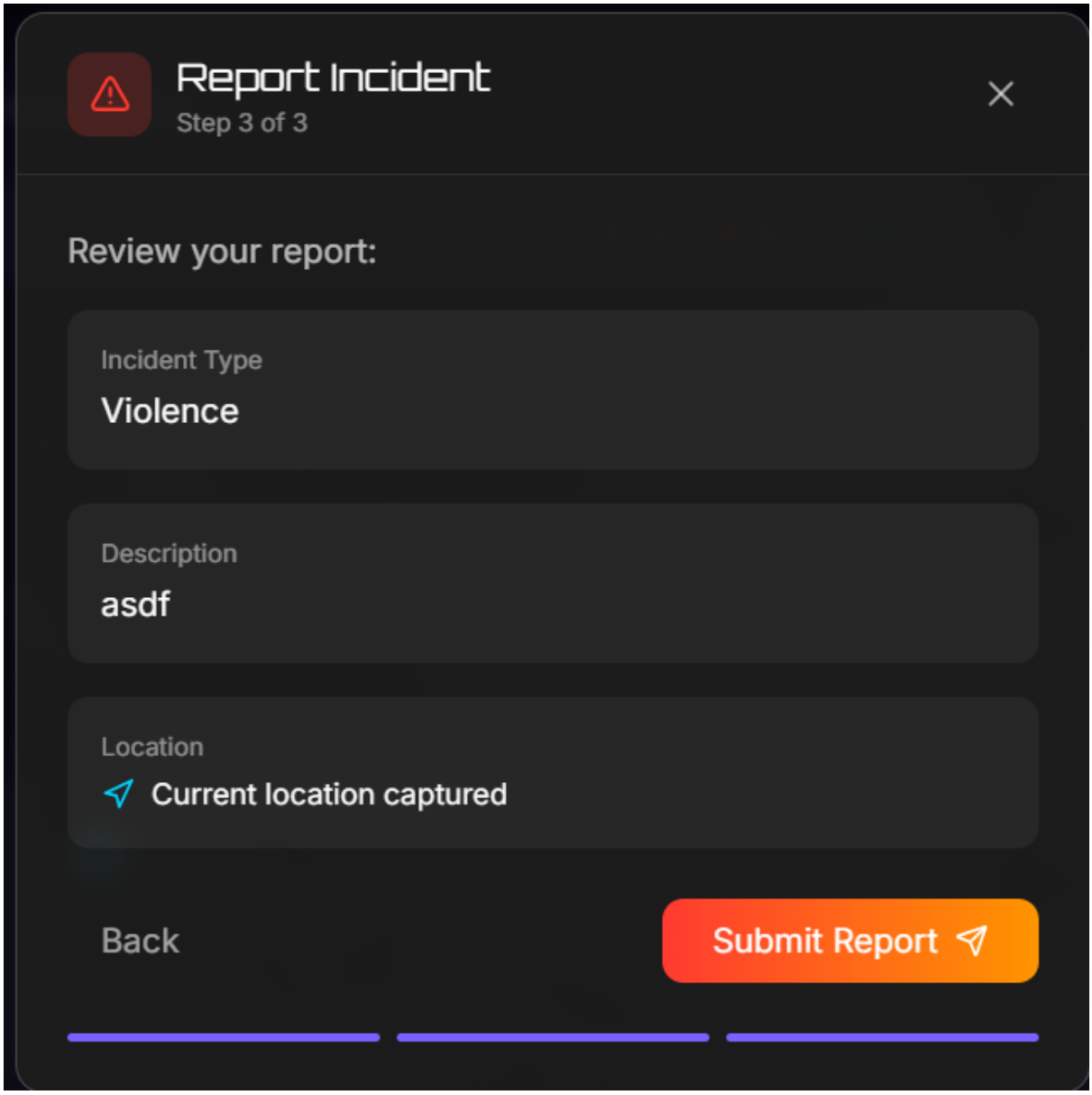
Report Incident

Step 1 of 3

Select incident type:

- Theft/Robbery
- Harassment
- Fire
- Medical Emergency
- Accident
- Violence
- Suspicious Activity
- Other


Report Incident
Step 2 of 3
Describe the incident:
What happened? Provide details...
Add Photo
Voice Note
Back
Next


Report Incident
Step 3 of 3
Review your report:
Incident Type
Violence
Description
asdf
Location
Current location captured
Back
Submit Report

MongoDB Compass - SHIELD/shield.users

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

SHIELD
- admin
- config
- local
- shield
  - incidents
  - users

users

SHIELD > shield > users

Open MongoDB shell

Documents 2 | Aggregations | Schema | Indexes 2 | Validation

Type a query: { field: 'value' } or Generate query

Explain | Reset | Find | Options

ADD DATA | UPDATE | DELETE | EXPORT DATA | EXPORT CODE

25 | 1 - 2 of 2


```
_id: ObjectId('69a5c777c377c2107757af9a')
name : "Krishna"
email : "abc@text.com"
password : "$2b$12$lP4S.RyamIAB.wBR0kc8CengmtTa7uRahw2ymEJRgynu61j37e4J5"
role : "citizen"
trustScore : 50
createdAt : 2026-03-02T17:23:03.718+00:00
__v : 0
```

```
_id: ObjectId('69a5c835c377c2107757afb5')
name : "Bandit"
email : "test@admin.com"
password : "$2b$12$ct4E.oSyA97GQluCYUACl.rMczuFySyKJJOVzdt6w3wbS.1.CwCnW"
role : "admin"
trustScore : 50
createdAt : 2026-03-02T17:26:13.290+00:00
__v : 0
```

6. Github repo link

https://github.com/krishnapriyadarshan2023-dev/D2_23BCE5138_Krishna-Priyadarshan